

# Daily Review

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Booklet p.15-17 — Forces on Rockets

- 1 State Newton's third law.
- 2 A 100 kg probe has an unbalanced force of 30 N. Find  $a$ .
- 3 Name the forces on a rocket at take-off.
- 4 What must the thrust be bigger than to lift off?
- 5 What is a Newton pair?

Every action has an equal, opposite reaction

$0.30 \text{ m s}^{-2}$

Thrust and weight

The weight

Action and reaction

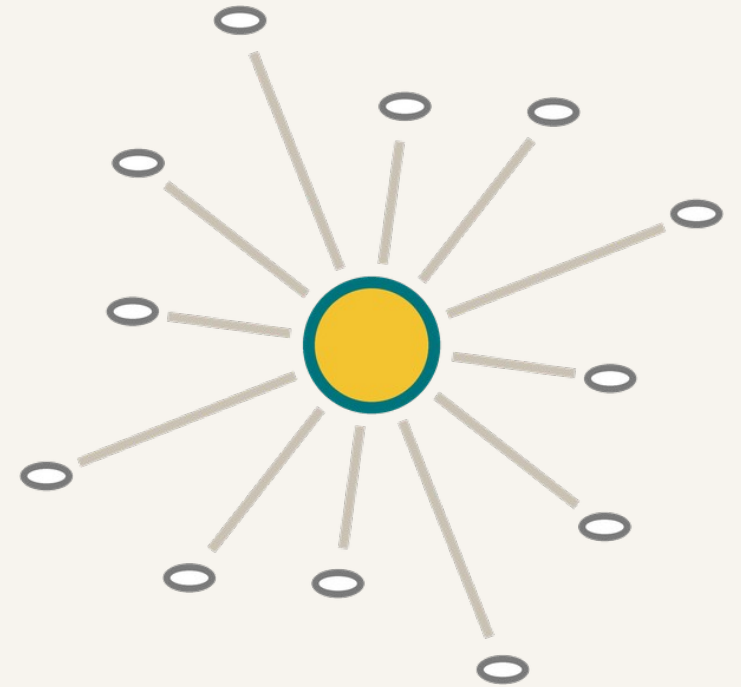
SPACE

# The Big Bang

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Where did everything come from —  
and when?

National 5 Physics (S3)



## THE BIG QUESTION

Almost every smudge here is a whole galaxy, and they are all rushing apart. What does that say about the start?



Webb, 2022 — NASA, ESA, CSA, STScI

PART 1

# The Big Bang Theory

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How the universe began, and how old it is.

## By the end of this part...

- 1 I can give a basic description of the Big Bang theory
- 2 I can state the estimated age of the universe

# Show me — Describe each word

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SHOW ME BOARDS

1 universe

Everything that exists

2 galaxy

A huge group of stars

3 the Big Bang

The theory of how the universe began

4 expanding

Getting bigger; space itself stretching

5 evidence

What we observe that supports a theory

# Show me — Describe each word

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SHOW ME BOARDS

6

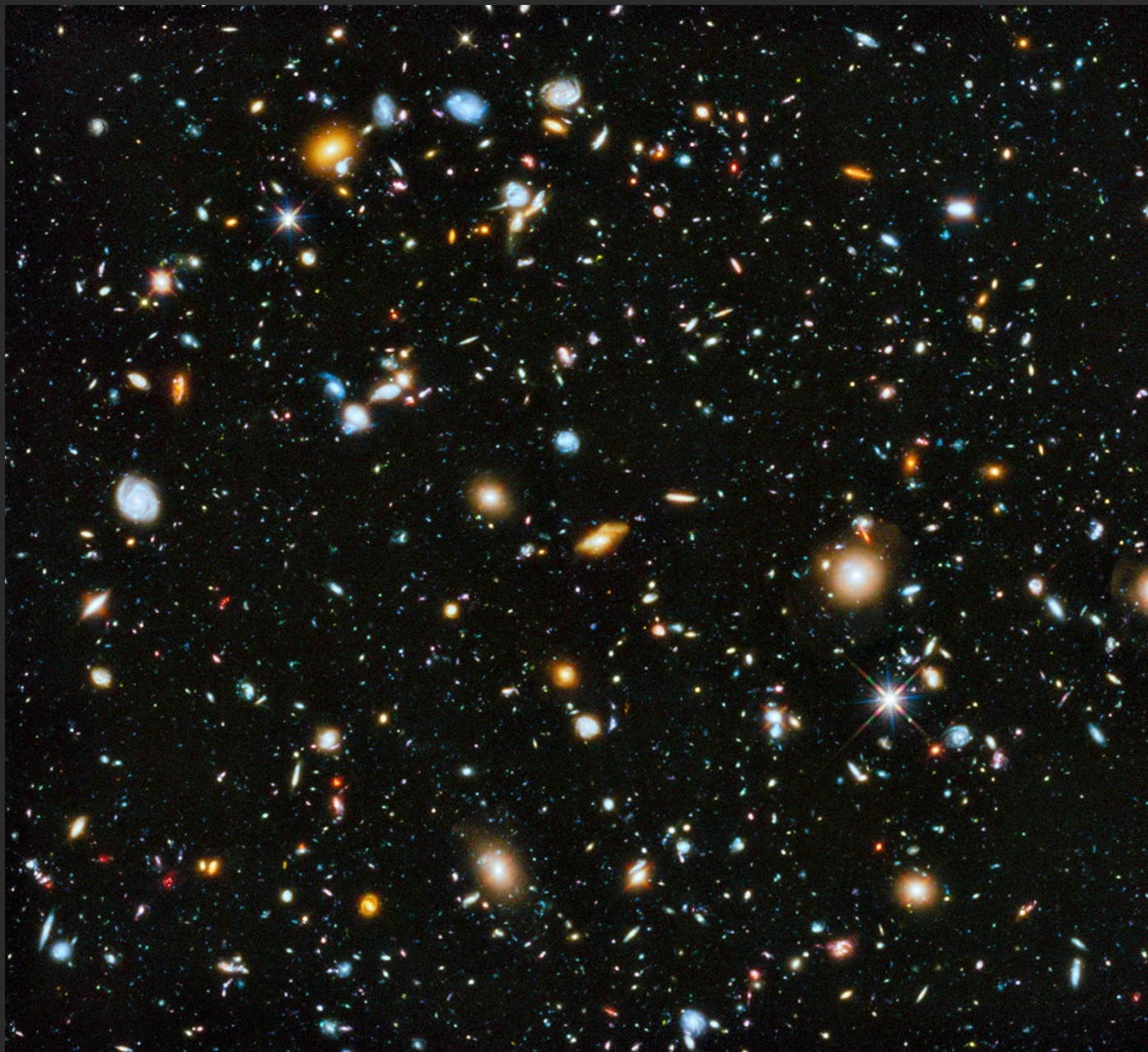
redshift

Light stretched to a longer wavelength

7

microwave background

The cooled glow of the early universe

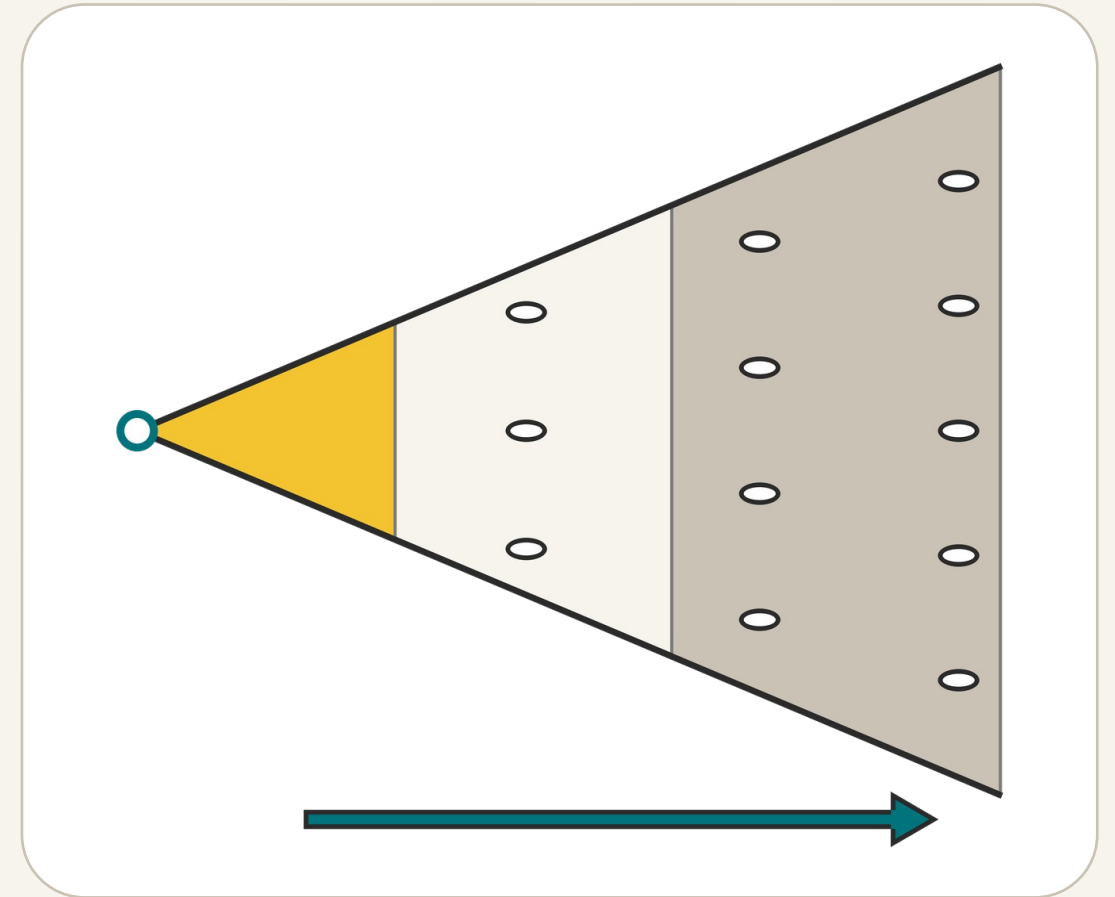


The Hubble Ultra Deep Field — NASA / ESA

# The Big Bang

Booklet p.20

- The universe began as a tiny, very hot, very dense point.
- Since then it has been expanding and cooling.
- Galaxies are still moving further apart today.



# How old is the universe?

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Booklet p.20

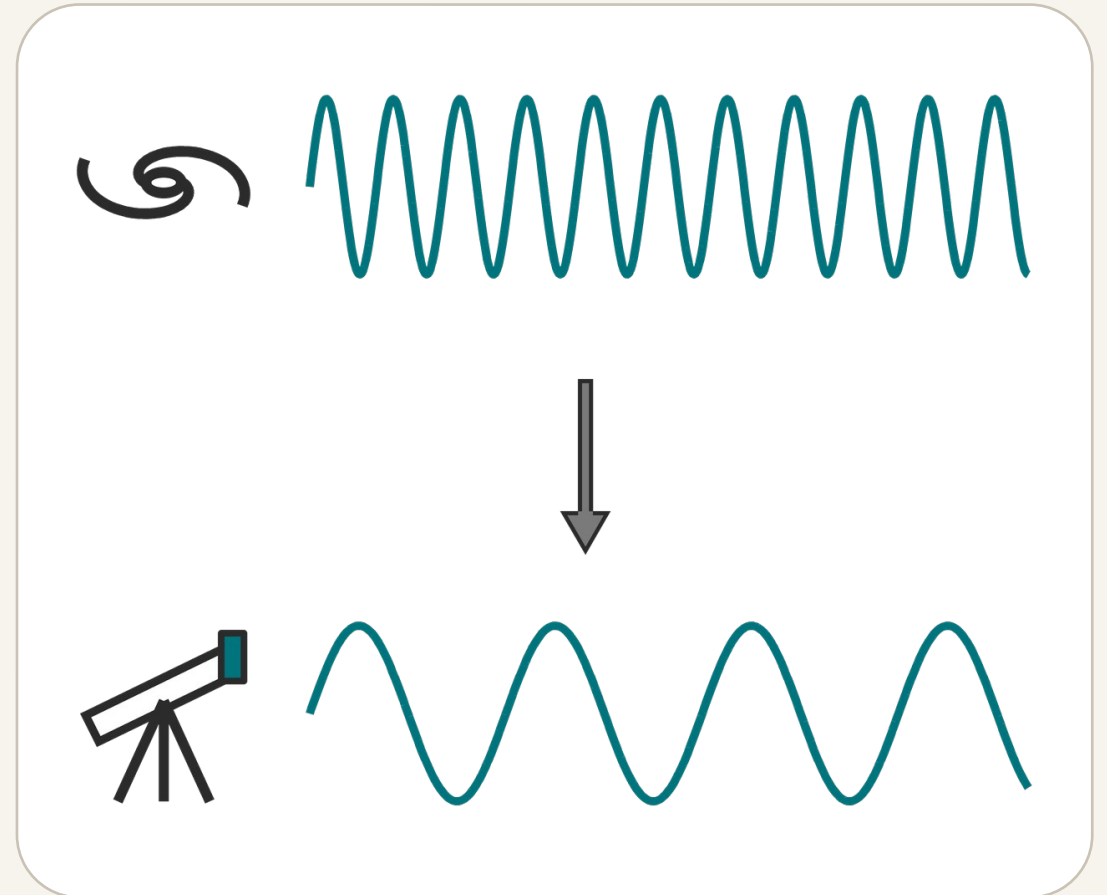
**About 13.8 billion years**

The estimated time from the Big Bang to today.

# Expanding space stretches light

## Evidence for the Big Bang

- Space itself expands: the gaps between galaxies grow.
- Light crossing those gaps is stretched with them.
- It arrives with a longer wavelength – redshifted.
- The further away a galaxy is, the redder its light.



# Hinge question

FINGER VOTE

Think first — then show A to E on your fingers, 1 to 5

Which of these best describes the Big Bang?

- A** An explosion of a star out in space
- B** Space itself expanding from a tiny hot point
- C** An explosion at the centre of the universe
- D** Matter flying out into empty space
- E** The Earth forming from a cloud of dust.

Answer B

# Show me — the Big Bang

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SHOW ME BOARDS

EXTRA PRACTICE

- 1 Is the universe getting hotter or cooler?
- 2 Give two words for the universe at its start.
- 3 Is 13.8 billion years more than 14 million?
- 4 What does the theory describe the origin of?
- 5 Was the universe bigger or smaller in the past?

Cooler

Hot and dense

Yes, far more

The universe

Smaller

# Agree, disagree or it depends?

SHOW ME BOARDS

Boards up: AGREE, DISAGREE or IT DEPENDS — and why

- 1 The Big Bang was an explosion in space.  
Disagree: space itself expanded
- 2 The universe is expanding into something.  
Disagree: there is no outside
- 3 Redshift means the light slows down.  
Disagree: it is stretched
- 4 Light from distant galaxies is redder.  
Agree: it has been stretched
- 5 The universe has a centre.  
Disagree: no centre

# Task 1 – the age of the universe

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Booklet p.20

Complete the definition.

**Booklet p.20, Task 1**

# Task 1 – Answers

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## ANSWERS

Booklet p.20

- The universe is approximately 13.8 billion years old.

## Task 2 – describe the Big Bang

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Booklet p.20

Watch the clip about the Big Bang,  
then describe the theory.

**Booklet p.20, Task 2**

## Task 2 – Answers

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### ANSWERS

Booklet p.20

- The universe began as a very small, very hot, very dense point.
- It has been expanding and cooling ever since.

# Extension – the evidence

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Booklet p.20

Research the three pieces of evidence  
and describe each one.

**Booklet p.20, Extension**

# Extension – Answers

## ANSWERS

Booklet p.20

| Evidence             | What it shows                                                 |
|----------------------|---------------------------------------------------------------|
| Microwave background | the cooled glow of the early universe, from every direction   |
| Olbers' paradox      | the night sky is dark, so the universe has not lasted forever |
| Light elements       | the amounts of hydrogen and helium match the Big Bang         |

# Checkpoint

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- 1 Give a basic description of the Big Bang theory.
- 2 State the estimated age of the universe.
- 3 What has the universe been doing since the Big Bang?

A tiny hot dense point,  
expanding ever since

About 13.8 billion years

Expanding and cooling

# Exam practice

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Past paper and specimen paper

- National 5 Physics 2024, Section 1, question 8. 1 mark.
- Three statements about the universe — which are correct?
- Specimen paper, Section 1, question 6. 1 mark. The same pattern.
- Then booklet p.21, Task 3.

LOOKING AHEAD

# Spectra

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Starlight carries a barcode. Reading it tells us what a star is made of.

# Plenary

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- 1 The big question: what does everything rushing apart tell us?
- 2 Describe the universe at the moment of the Big Bang.
- 3 What has happened to the universe since?
- 4 State the estimated age of the universe.
- 5 Name one piece of evidence for the Big Bang.

It was all together once

Very small, very hot,  
very dense

It has expanded and  
cooled

About 13.8 billion years

The microwave  
background

# Plenary

EXTRA PRACTICE

- 1 Is the universe expanding or shrinking?  
Expanding
- 2 Why does light from a distant galaxy look redder?  
Space stretched it on the way
- 3 Is the universe hotter or cooler now than at the start?  
Cooler
- 4 Were galaxies closer together in the past?  
Yes
- 5 What does the Big Bang theory describe?  
The origin of the universe

# Extension

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- 1 Why is 'an explosion in space' a misleading picture of the Big Bang?
- 2 Suggest why the Big Bang is called a theory.
- 3 Why can we not see anything older than the universe itself?

# Extension

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## EXTRA PRACTICE

- 1 If galaxies are moving apart now, what must have been true in the past?
- 2 The microwave background comes from every direction. Why is that important?
- 3 Suggest why the night sky would be bright in a never-changing universe.

# Course specification

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## N5 Physics course specification v4.0, p.8 – Cosmology

- Basic description of the ‘Big Bang’ theory of the origin of the universe.
- Knowledge of the approximate estimated age of the universe.